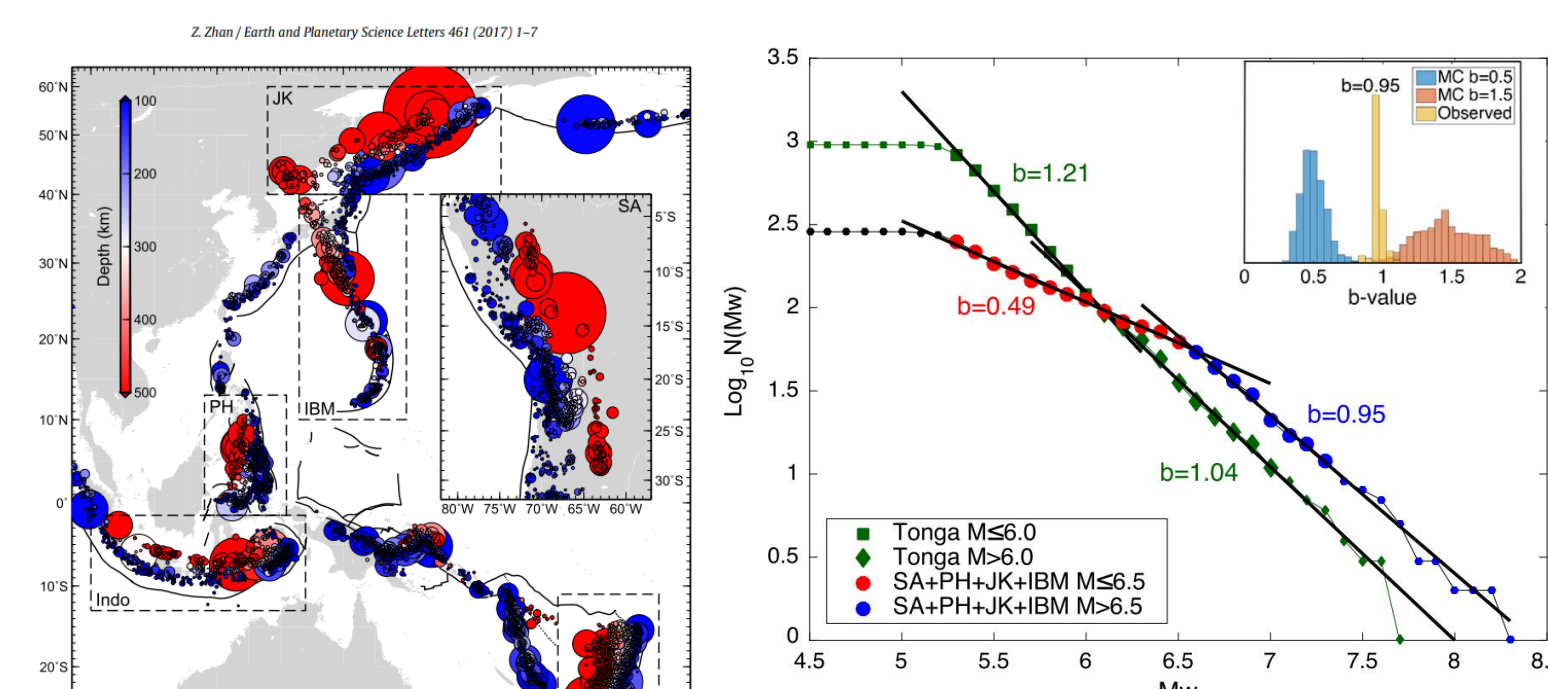
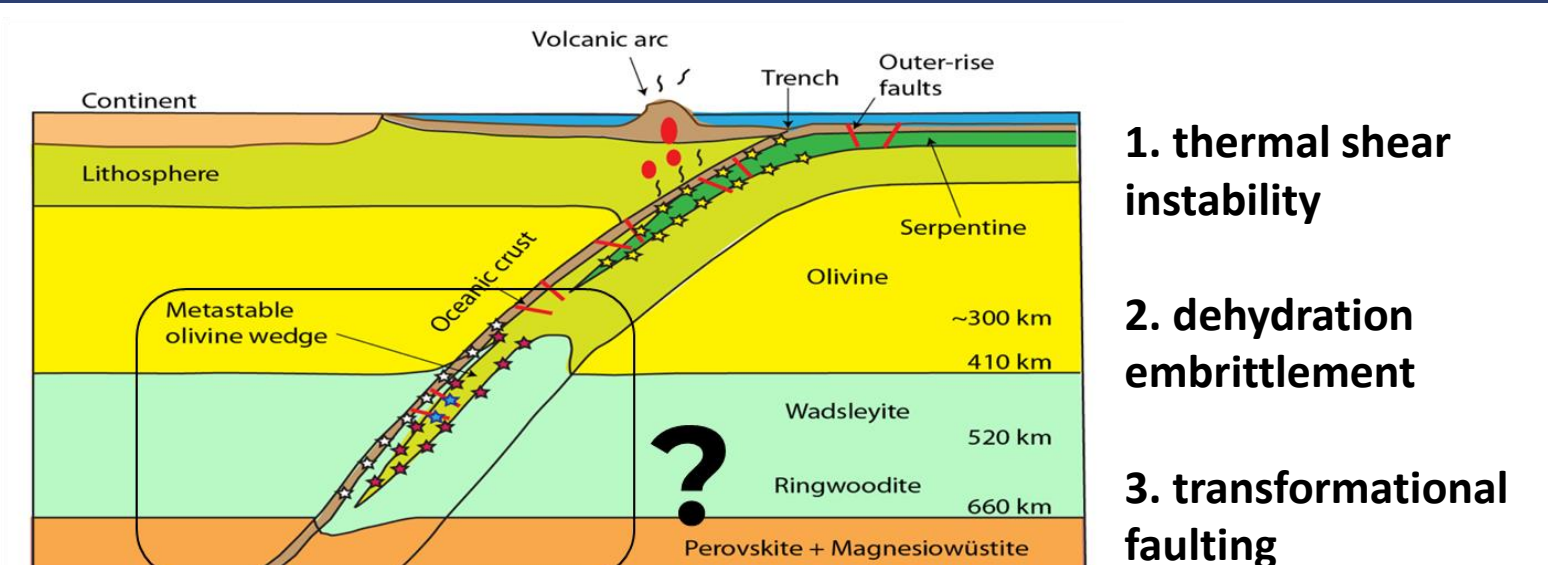


# Nature of Deep Earthquakes in the Pacific Plate from Unsupervised Machine Learning

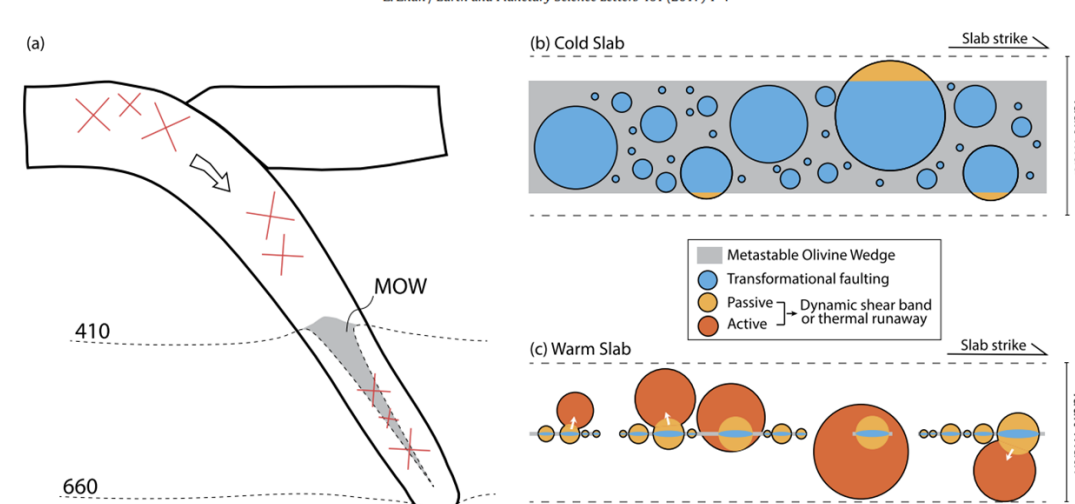
Gilbert Mao<sup>1,2</sup>, Thomas P. Ferrand<sup>3,4</sup>, Jiaqi Li<sup>1,5</sup>, Brian Zhu<sup>1,6</sup>, Ziyi Xi<sup>1</sup> and Min Chen<sup>1,7</sup>

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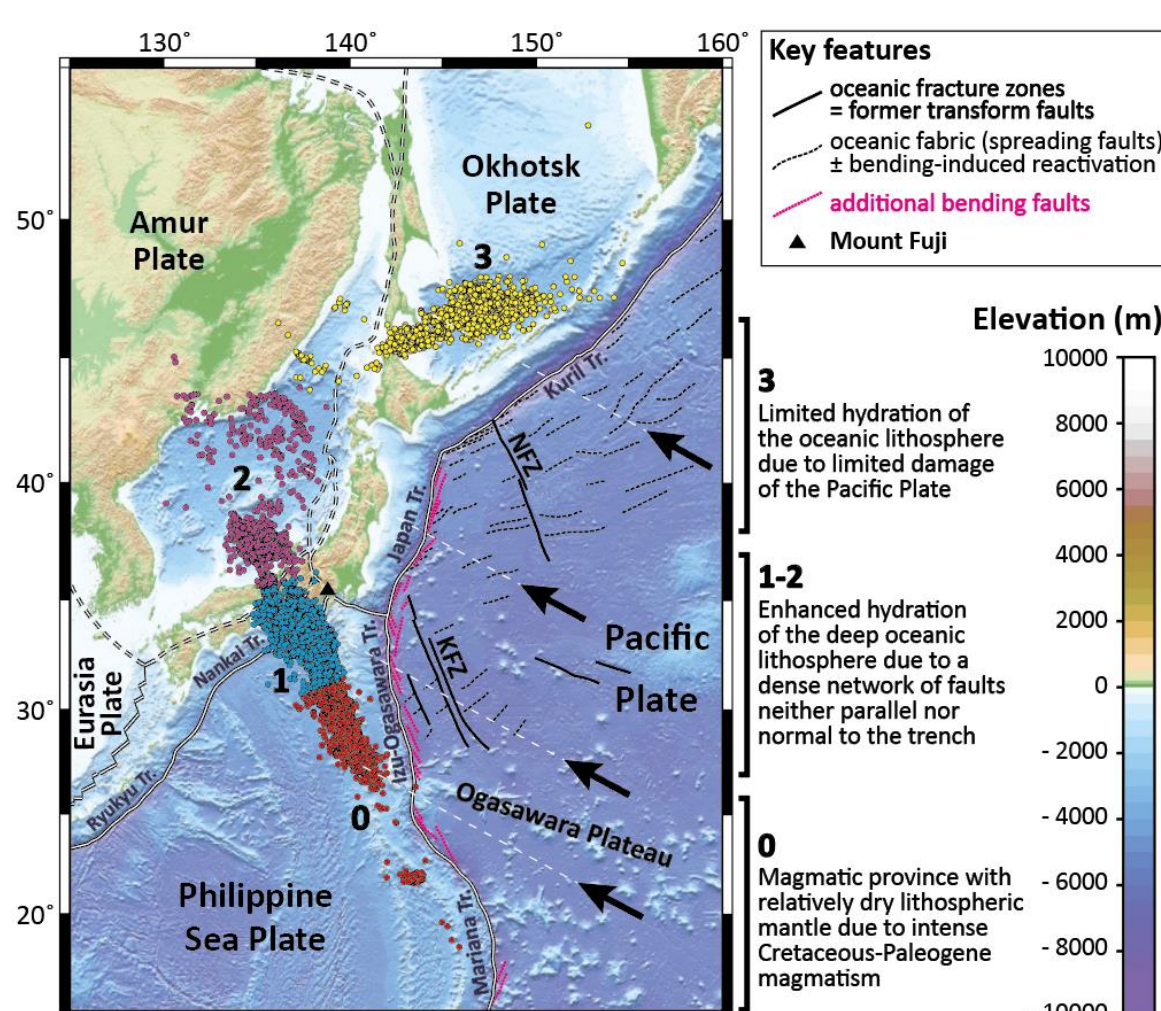
## INTRODUCTION



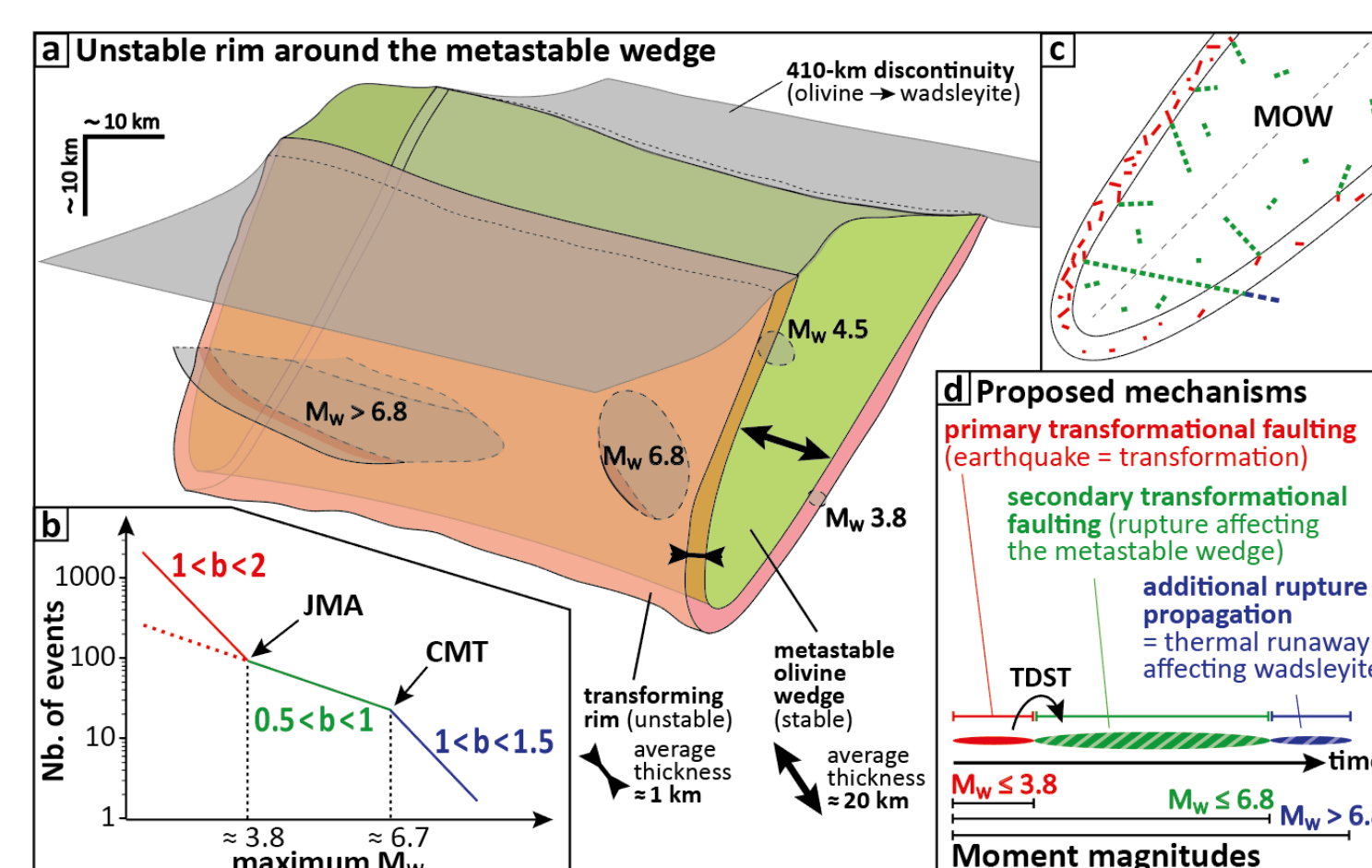
### Dual Mechanism Hypothesis Based on b-value Analysis of CMT Catalog



## Mechanisms for the b-value Kinks

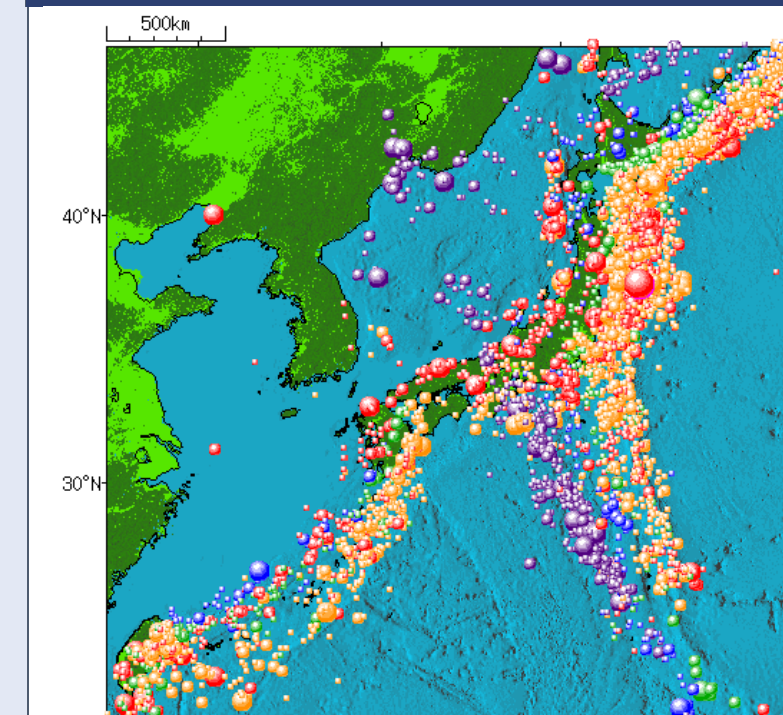


Pre-Subducting Plate Hydration Affects the Water Abundance in the MOW Halo.



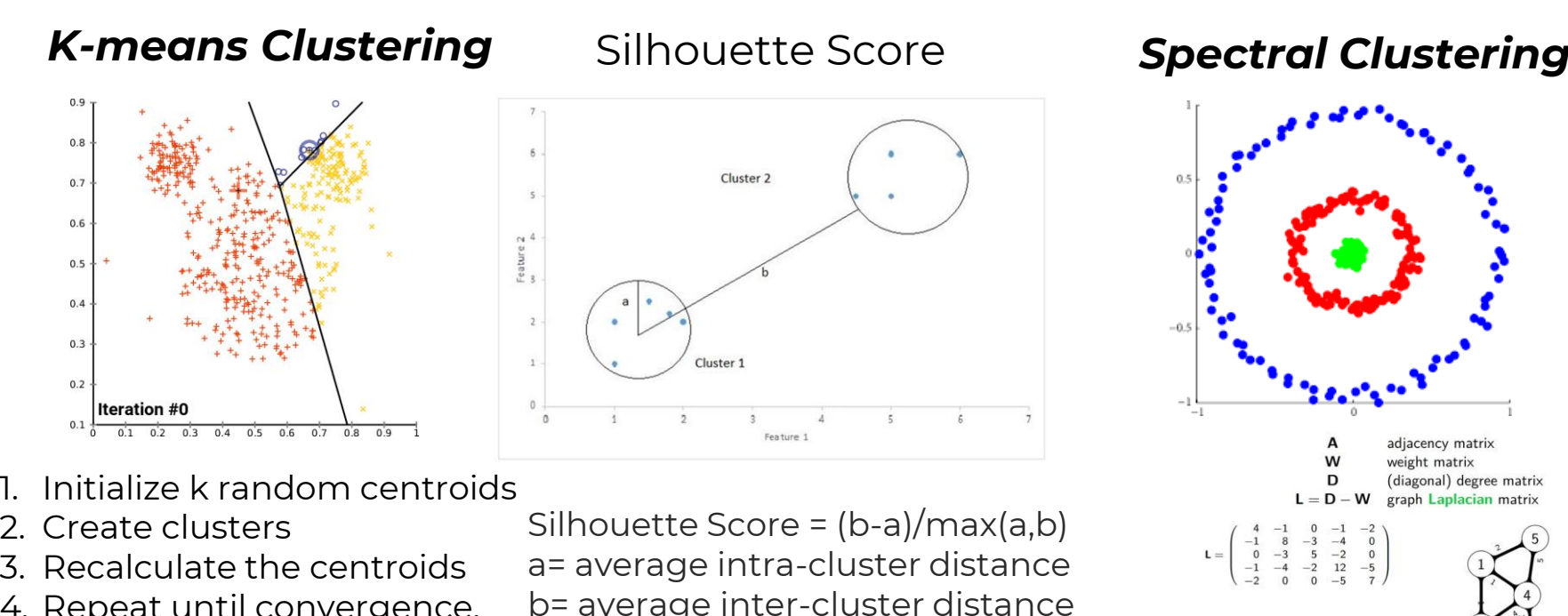
Deep-Focus Earthquakes Triggered in the Unstable Transforming Rim and can Propagate Into and then Outside the MOW. Water Acts as a Catalyst of Small Earthquakes in a Highly Reactive Halo of MOW.

## Data & Analysis



- More complete earthquake catalog for Japan and its surrounding regions
- JMA magnitude (Mj) has been converted to Mw (Uchide & Imanishi, 2018)  
 $M_w = aM_j^2 + bM_j + c$   
 $a = 0.053 \pm 0.003$ ,  $b = 0.33 \pm 0.02$ , and  $c = 1.68 \pm 0.03$
- JMA Earthquake Magnitude of Completeness ( $M_c \sim 3.3$ ) much lower than CMT catalog ( $M_c \sim 5$ )
- Over two million earthquakes

### Clustering Analysis with Unsupervised Machine Learning:



- Initialize k random centroids
- Create clusters
- Recalculate the centroids
- Repeat until convergence.

### b-value analysis and its error analysis:

$$b_0 = \frac{1}{\ln(10) * (\mu - M_c)}$$

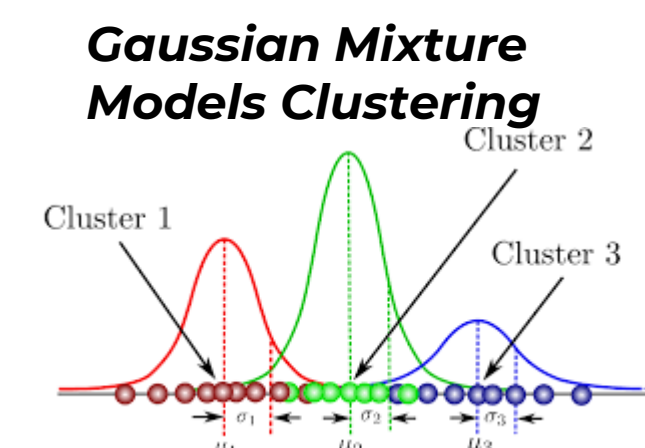
$$b_1 = \frac{1}{\ln(10) * (\mu - (M_c - \Delta M/2))}$$

$$b_2 = \frac{1}{\ln(10) * \Delta M * \ln(1 + \frac{\Delta M}{\mu - M_c})}$$

$$\sigma_{b_0} = 2.3 b_0^2 \sqrt{\frac{\sum_{i=1}^N (M_i - \mu)^2}{N(N-1)}}$$

$$\sigma_{b_1} = 2.3 b_1^2 \sqrt{\frac{\sum_{i=1}^N (M_i - \mu)^2}{N(N-1)}}$$

$$\sigma_{b_2} = \frac{\Delta M}{\ln(10) * \Delta M * \sqrt{N * (1 + \frac{\Delta M}{\mu - M_c})}}$$

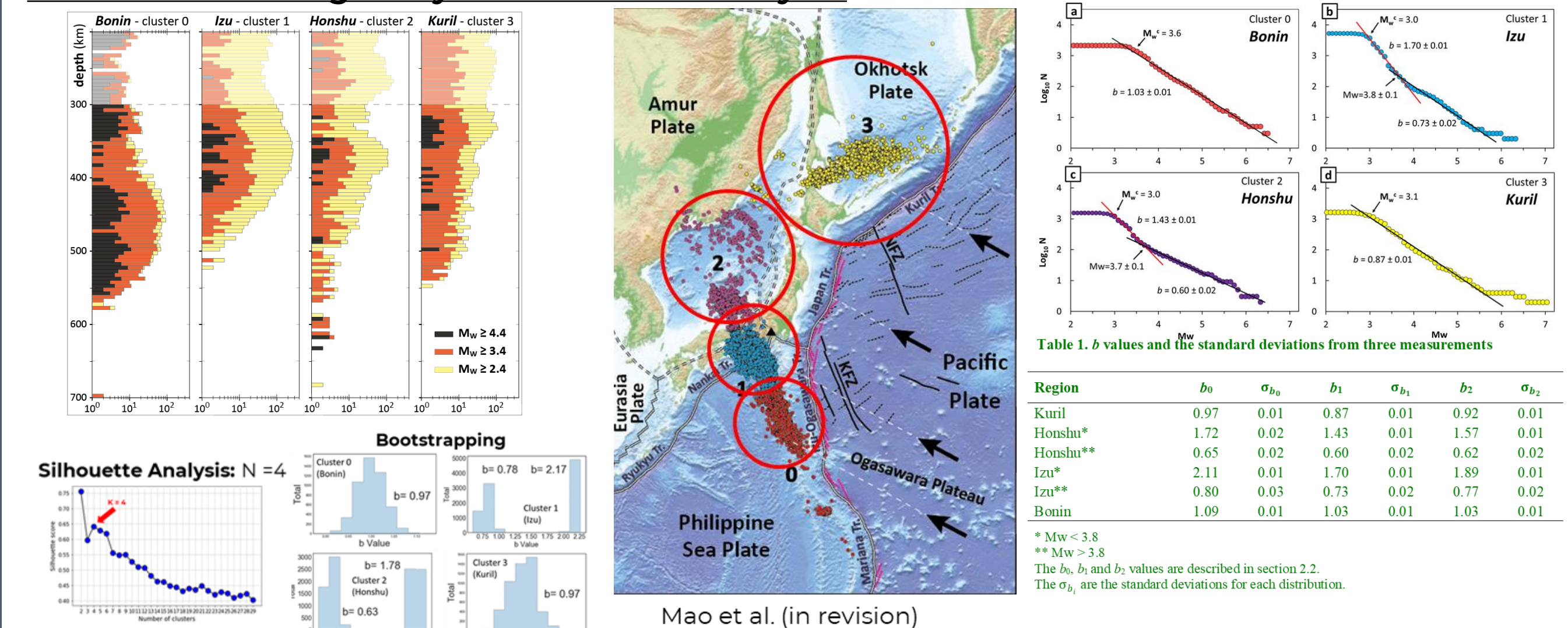


## References

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- Ishii, T. & Ohtani, E. Dry metastable olivine and slab deformation in a wet subducting slab. *Nature Geoscience* **14**, 526–530 (2021).
- Warner, M. & Sandri, L. A review and new insights on the estimation of the b-value and its uncertainty. *Annals of Geophysics* **46**(6) (2003).
- Ferrand, T. P. *et al.* Dehydration-driven stress transfer triggers intermediate-depth earthquakes. *Nature Communication* **8**, 1–11 (2017).

## Results & Conclusion

### K-means Clustering Analysis and b-value Analysis:



Mao et al. (in revision)

### Spectral and GMM Clustering Analysis:

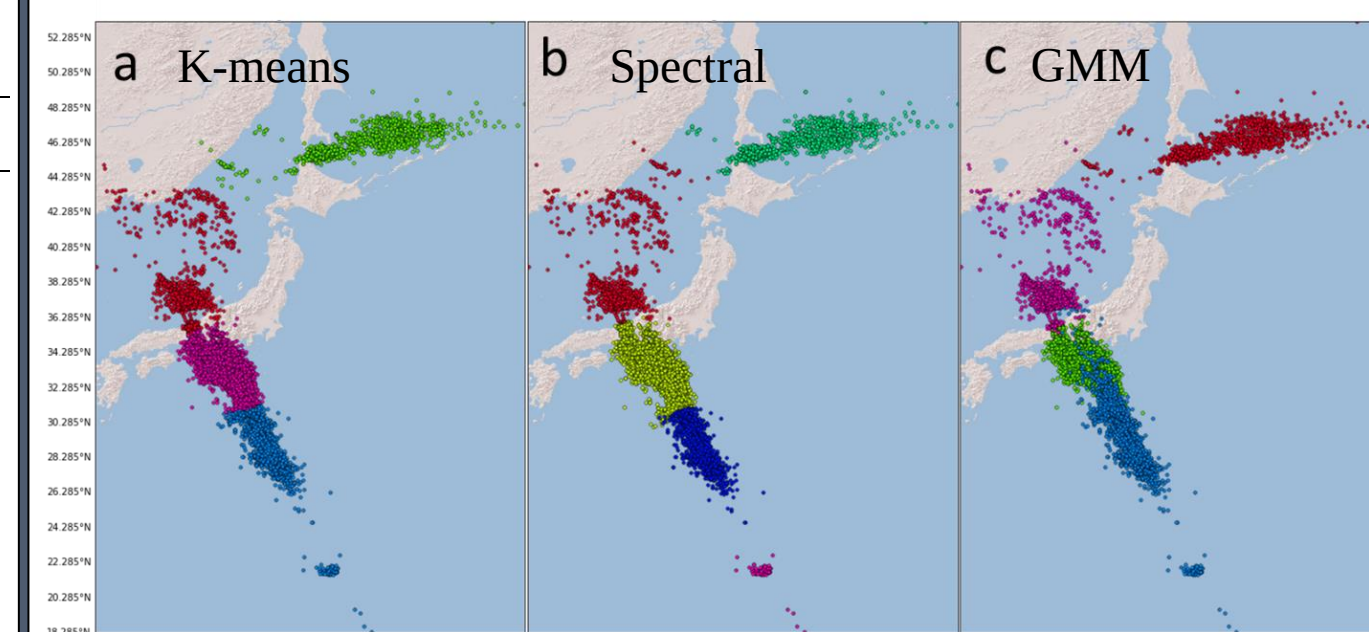
Table 2. b values and their errors from three measurements (Spectral Clustering, N=5)

| Region   | $b_0$ | $\sigma_{b_0}$ | $b_1$ | $\sigma_{b_1}$ | $b_2$ | $\sigma_{b_2}$ |
|----------|-------|----------------|-------|----------------|-------|----------------|
| Kuril    | 0.97  | 0.01           | 0.87  | 0.01           | 1.00  | 0.01           |
| Honshu*  | 1.70  | 0.02           | 1.42  | 0.01           | 1.43  | 0.01           |
| Honshu** | 0.66  | 0.02           | 0.61  | 0.02           | 0.61  | 0.02           |
| Izu*     | 2.10  | 0.01           | 1.69  | 0.01           | 1.71  | 0.01           |
| Izu**    | 0.79  | 0.02           | 0.72  | 0.02           | 0.73  | 0.02           |
| Bonin    | 0.93  | 0.01           | 0.84  | 0.01           | 0.95  | 0.01           |

Table 3. b values and their errors (Gaussian Mixture Models Clustering, N=4)

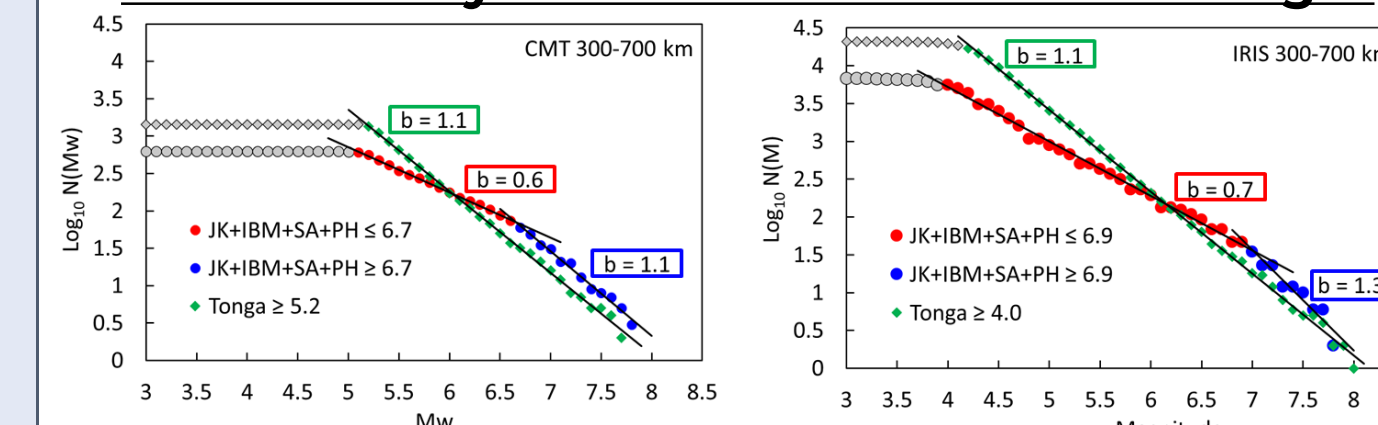
| Region   | $b_0$ | $\sigma_{b_0}$ | $b_1$ | $\sigma_{b_1}$ | $b_2$ | $\sigma_{b_2}$ |
|----------|-------|----------------|-------|----------------|-------|----------------|
| Kuril    | 0.97  | 0.01           | 0.87  | 0.01           | 0.88  | 0.01           |
| Honshu*  | 1.70  | 0.02           | 1.42  | 0.01           | 1.43  | 0.02           |
| Honshu** | 0.65  | 0.02           | 0.61  | 0.02           | 0.61  | 0.02           |
| Izu*     | 2.17  | 0.01           | 1.74  | 0.01           | 1.76  | 0.01           |
| Izu**    | 0.83  | 0.03           | 0.76  | 0.02           | 0.76  | 0.03           |
| Bonin    | 1.09  | 0.01           | 0.97  | 0.01           | 0.97  | 0.01           |

### Comparison of Three Clustering Methods:



Three clustering methods generated similar clustering maps. The b-value analysis results are also similar. The clustering analysis of the JMA catalog is robust.

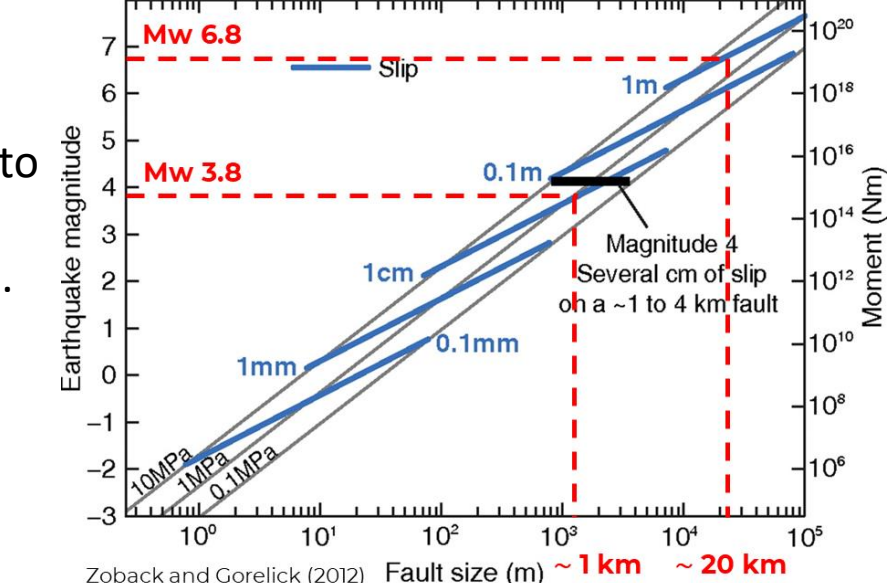
### b-value analysis with CMT and IRIS catalogs:



- IRIS and CMT catalogs show similar earthquake distributions
- Warm subduction zones show a change of b-values at Mw~6.8
- Cold Tonga subduction zone has a constant b-value of ~1.1

In JMA catalog, a new kink at Mw 3.8 was observed, in addition to the kink at Mw 6.5 in CMT and IRIS catalogs.

Two b-value kinks indicate two different rupture dimension thresholds.



## Conclusion:

- Unsupervised machine learning approaches facilitate the unbiased clustering of deep-focus earthquakes from the JMA catalog.
- The b-value kink at Mw~3.8 provides strong evidence for phase transformational faulting triggering deep-focus earthquakes in an unstable rim of the metastable olivine wedge with a width of ~1 km.
- In the highly reactive halo, water acts as a catalyst for transformational faulting, resulting in higher b-values in the Honshu and Izu slabs for Mw < 3.8.
- We speculate that the b-value kink at Mw~3.8 can be observed in other subduction zones with similar tectonic environments and earthquake catalogs.